

ARCHITECTURE COMPARISON

WHAT CHANGES WHEN IGNITION GROWS PAST A FEW SITES

Five sites can be run almost any way. Forty cannot. This page compares three ways to build a distributed Ignition estate on the problems that appear with scale: fan-in, backend count, outages, timestamps, adding a consumer, and running the fleet afterwards. It is written for the engineer who will own the result, and it names what each approach does well.

THE THREE SHAPES

TRADITIONAL, MODULE-BASED MQTT, SPARKBRIDGE

The same forty sites, drawn three ways. Every box is a server somebody has to license, patch and watch.

TRADITIONAL DISTRIBUTED IGNITION

```

40 site gateways
  |
Gateway Network, remote tag providers
  |
Backend 1 ---- Backend 2 ---- Backend 3
  |           |           |
History      Applications Integrations
(per backend) (per backend) (per system)
  
```

[● ASK AN ENGINEER](#)

Every consumer is another Gateway Network connection into every site, or another backend that holds its own copy of the estate. Site count multiplies backend load; consumer count multiplies site load.

SPARKBRIDGE

```
40 x Edge (publish once, buffer to disk)
|
Provider HA pair (identity, policy, routing)
|
Host pool (one namespace, shared consumers)
|
+-----+-----+-----+-----+
Vault  Record  Passage  Inject  Your tools
```

Sites publish once. Consumers subscribe at Host. Adding a consumer, a site, or a Host changes one layer and leaves the other two alone.

Module-based MQTT for Ignition sits between the two: sites publish over Sparkplug and buffer through outages, which removes polling and most of the fan-in. What it leaves to custom engineering is everything after the transport: consumer scale-out, provisioning, configuration drift, fleet health and security policy. MQTT is the transport. Operating the estate is the problem.

PROBLEM BY PROBLEM

WHERE EACH ARCHITECTURE SPENDS YOUR ENGINEERING TIME

Rows are the problems that arrive with site count. Cells say who solves them: the product, or your team.



PROBLEM AT SCALE	TRADITIONAL DISTRIBUTED IGNITION	MODULE-BASED MQTT	SPARKBRIDGE
Gateway Network fan-in	One hub gateway carries every site connection and is the first thing to saturate	Removed: sites publish to a broker	Removed: sites publish to Provider; Host consumers share the stream
Remote tag provider scaling	Each remote provider is a live subscription load on the site	Removed at the site; consumer side is the engine's tag count	Host holds 500,000 wire metrics in under 4 GiB; SparkFlow streams past the tag ceiling
Backend server proliferation	A backend per application family, each with its own copy of the estate	One engine gateway; a second consumer usually means a second engine	One Host pool; consumers are modules on Host or subscribers of the same stream
Database connection sprawl	Each backend writes its own history and reads its own DB	History still written per gateway that historizes	SparkVault writes once, centrally, on source timestamps; SparkInject lands the warehouse copy exactly once
Historian duplication	Site history plus central history, reconciled by hand	Depends on where history is enabled	History is one consumer of the backbone. Adding it changes nothing at the sites



PROBLEM AT SCALE	TRADITIONAL DISTRIBUTED IGNITION	MODULE-BASED MQTT	SPARKBRIDGE
WAN outages	Gateway Network reconnects; history during the gap depends on site-side store-and-forward configuration	Store-and-forward at the site, replayed on reconnect	Disk-backed buffer at 2,697 messages/s durable, replayed in order, 400,000 of 400,000 measured, with the historical flag carried so trends backfill
Timestamp integrity	Arrival time unless every hop is configured to preserve source time	Source timestamps carried by Sparkplug	Source timestamps carried through buffer, replay, history and warehouse delivery. A timestamp that survives the trip
Consumer scaling	Another backend, another set of site connections	Another subscriber on the broker, one wildcard subscriber per engine	MQTT 5 shared subscriptions: four Host consumers deliver about 20x one. Publish once, add consumers when you need them
Site provisioning	Manual gateway build, GAN approval, provider setup per site	Broker URL, credentials and certificates typed at each site	One-time enrollment code: the site bootstraps identity over a signed channel, credentials written owner-only. Certificate issuance and fleet registration are on the roadmap
Configuration drift	Per-gateway settings, reviewed by visiting each one	Per-gateway module settings	Signed configuration bundles distributed fleet-wide; FleetOps shows drift per gateway; GitOps / Enterprise Governance (pilot) adds review and rollback

PROBLEM AT SCALE	TRADITIONAL DISTRIBUTED IGNITION	MODULE-BASED MQTT	SPARKBRIDGE
Fleet operations	Gateway status pages, one at a time	Broker client list plus gateway status pages	FleetOps: versions, drift, certificates, backups per gateway in one view; Host /fleet and Prometheus metrics
Security policy	Gateway Network certificates and per-gateway users	Broker ACL files written by hand	SparkID generates deny-by-default rules from intent; Provider enforces per-site identities; Sentinel scores the result

Where a module-based MQTT cell reads the same as SparkBridge, that is the Sparkplug standard doing its job for both. The difference is what remains for your team to build afterwards.

5 TO 50

SCALING FROM FIVE SITES TO FIFTY

What each step costs in infrastructure, not licenses. License arithmetic is on [pricing](#).

SITE COUNT	TRADITIONAL DISTRIBUTED IGNITION	SPARKBRIDGE
5	One hub gateway, remote tag providers, fine	5 Edge, 1 Provider, 1 Host. Also fine
20	Hub CPU and memory climb with every remote provider; history moves to a second backend	Same three roles. Host memory grows about 300 bytes per wire metric per consumer

SITE COUNT	TRADITIONAL DISTRIBUTED IGNITION	SPARKBRIDGE
40	Hub is the overloaded box; a second hub splits the estate and every consumer now needs both	Provider HA pair, Host active plus warm standby (takeover measured at 244 ms across 25 nodes). One namespace
50 and beyond	Backend per consumer family; integration projects per new tool	Add Host consumers on the shared stream; add SparkFlow when tag count is the ceiling. Sites are not touched

ADD A CONSUMER

ADDING A CONSUMER WITHOUT TOUCHING FORTY PLANTS

DAY ONE

40 sites --> Provider --> Host --> Existing application

A YEAR LATER

40 sites --> Provider --> Host --> Existing application
--> SparkVault (history)
--> SparkInject (warehouse)
--> Passage (OPC-UA clients)
--> Analytics and AI

No changes at the forty sites. Each new consumer is a subscription at Host or a module on it. In the traditional shape each of those four lines is a new backend or a new set of site connections, and usually a project with a budget line.



OUTAGE

WAN OUTAGE RECOVERY, MEASURED

MOMENT	WHAT SPARKBRIDGE DOES	MEASURED
Link drops	Edge diverts to a disk buffer; Host marks the site's points bad, not stale-but-current	2,697 messages/s durable with batched fsync
Link returns	Edge reconnects within a bounded backoff, rebirths, replays in order with source timestamps and the historical flag	Reconnect capped at 10 s; 400,000 of 400,000 replayed, zero loss, 40,035 updates/s on the reference rig
Host fails	Warm standby promotes after a debounce; edges never rebirth	244 ms from offline STATE to online STATE across 25 nodes
Afterwards	SparkVault backfills the real curve; SparkInject lands each row once	Zero duplicate or missing rows in the record lane soak

Environment and method for every figure: [Scale and Resilience Report](#).

RUNNING IT

FLEET OPERATIONS AFTER GO-LIVE

The questions that arrive in month two, and where the answer lives.



Which sites are offline or buffering?

Host /fleet view with store-and-forward and capacity headroom per edge; Prometheus metrics

Which versions are deployed and which drifted?

FleetOps inventory and drift, per gateway

Which certificates expire soon?

FleetOps expiring list; Sentinel finding with the fix

Can site A publish as site B?

No by construction: Provider binds each credential to one group and node. Sentinel proves it

How much headroom is left?

Host /capacity reports its own ceiling from benchmark evidence and the shard count it runs

Who changed what?

Signed configuration bundles today; review, rollback and audit trail in GitOps / Enterprise Governance (pilot)

COST

ARCHITECTURE COST, COMPARED HONESTLY

Ignition gateway licenses are the same in every shape and are not counted here. The comparison is what sits on top of them.

ITEM

TRADITIONAL DISTRIBUTED IGNITION

SPARKBRIDGE, 40 SITES

Central servers
to run

Hub plus one backend per
application family, commonly
three or more

Provider pair and Host pair, four small
servers, or two with both roles



ITEM	TRADITIONAL DISTRIBUTED IGNITION	SPARKBRIDGE, 40 SITES
Software on top of Ignition	None, or per-site historian licenses	\$40,300 on your own broker, \$49,780 with Provider HA and SparkVault, list, once. Detail
Adding site 41	Gateway build, GAN approval, provider setup, history setup	\$995 and an enrollment code
Adding a warehouse feed	An integration project per backend	SparkInject, \$2,995 per connector family, no site changes
Support	Your integrator's rate	Optional, 20% of list per year, direct to the engineers who built it
Metering	None	None. No per-tag, per-device or per-client counts

Module-based MQTT list prices are not reproduced here because the vendors publish them through resellers and they change. Bring the quote you hold and we will put it beside these numbers for your site count.

DRAW YOUR ESTATE BEFORE YOU BUY ANYTHING

Send the topology and we return a diagram, the failure domains and the sizing. It costs nothing and it is how most conversations with us start.

REQUEST AN ARCHITECTURE REVIEW

READ THE BENCHMARK REPORT



DOWNLOAD THE 40-SITE REFERENCE ARCHITECTURE (PDF)

SparkBridge · Sparkplug B / 3.0 · MQTT 3.1.1 and MQTT 5 · Ignition 8.1.19+

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